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Neural network emulation of the human ventricular cardiomyocyte action potential: a tool for more efficient computation in pharmacological studies

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Abstract

Computer models of the human ventricular cardiomyocyte action potential (AP) have reached a level of detail and maturity that has led to an increasing number of applications in the pharmaceutical sector. However, interfacing the models with experimental data can become a significant computational burden. To mitigate the computational burden, the present study introduces a neural network (NN) that emulates the AP for given maximum conductances of selected ion channels, pumps, and exchangers. Its applicability in pharmacological studies was tested on synthetic and experimental data. The NN emulator enabled a massive speed-up of more than 10^4 compared to regular simulations and the forward problem (find drugged AP for pharmacological parameters defined as scaling factors of control maximum conductances) on synthetic data could be solved with average root-mean-square errors (RMSE) of 0.47 mV in normal APs and of 13.6 mV in abnormal APs exhibiting early afterdepolarizations (72% of the emulated APs were aligning with the abnormality, and the substantial majority of the remaining APs demonstrated pronounced proximity). This demonstrates not only very fast and mostly very accurate AP emulations but also the capability of accounting for discontinuities, a major advantage over existing emulation strategies. Furthermore, the inverse problem (find pharmacological parameters for control and drugged APs through optimization) on synthetic data could be solved with high accuracy shown by a maximum RMSE of 0.21 in the estimated pharmacological parameters. However, notable mismatches were observed between pharmacological parameters estimated from experimental data and distributions obtained from the Comprehensive in Vitro Proarrhythmia Assay initiative. This reveals larger inaccuracies which can be attributed particularly to the fact that small tissue preparations were studied while the emulator was trained on single cardiomyocyte data. Overall, our study highlights

the potential of NN emulators as powerful tool for an increased efficiency in future quantitative systems pharmacology studies.

eLife assessment

This **valuable** study reports on the potential of neural networks to emulate simulations of human ventricular cardiomyocyte action potentials for various ion channel parameters with the advantage of saving simulation time in certain conditions. The evidence supporting the claims of the authors is **solid**, although the inclusion of open analysis of drop-off accuracy and validation of the neural network emulators against experimental data would have strengthened the study. The work will be of interest to scientists working in cardiac simulation and quantitative pharmacology.

1 Introduction

Computer models of human physiology are becoming increasingly detailed and mature and the area of ventricular cardiomyocyte electrophysiology (EP) is one of the most advanced. The most updated models include fine representations of ion movements through various important channels, pumps, and exchangers, and take the complex handling of intracellular calcium accurately into account [Grandi et al., 2010 [↗](#), O'Hara et al., 2011 [↗](#), Tomek et al., 2019 [↗](#), Bartolucci et al., 2020 [↗](#)]. While these models have individual strengths and limitations in replicating different aspects of physiology, pathology, and pharmacology [Corrado et al., 2020 [↗](#), Amuzescu et al., 2021 [↗](#)], their degree of credibility has reached a level that has led to an increasing number of applications in academia and beyond. This holds in particular for the pharmaceutical sector, where much effort is spent on using computer modeling to reduce traditional preclinical and clinical methodologies for assessing the efficacy and safety of novel drug candidates [Mirams et al., 2011 [↗](#), Passini et al., 2017 [↗](#), Li et al., 2019 [↗](#), Passini et al., 2021 [↗](#)]. To improve the regulatory assessment of a drug's proarrhythmic potential, the Comprehensive in Vitro Proarrhythmia Assay (CiPA) was proposed in 2013 following a workshop at the US Food and Drug Administration [Sager et al., 2014 [↗](#), Colatsky et al., 2016 [↗](#), Strauss et al., 2021 [↗](#)]. A central component is a computer model of human ventricular cardiomyocyte EP that is coupled to a pharmacological model describing the interaction between a given drug and multiple arrhythmia-relevant channels [Dutta et al., 2017 [↗](#), Li et al., 2017 [↗](#), Li et al., 2019 [↗](#)]. For a given drug, experimental channel block data are collected to inform the pharmacological model and corresponding simulations of the action potentials (AP; time course of the transmembrane potential) are performed to predict the proarrhythmic risk based on a mechanistically motivated biomarker [Chang et al., 2017 [↗](#), Li et al., 2017 [↗](#), Li et al., 2019 [↗](#)]. The prediction is then compared with findings in experimental [Blinova et al., 2018 [↗](#)] and clinical [Vicente et al., 2017 [↗](#)] studies. To compute the drugged AP for given pharmacological parameters is a forward problem, while the corresponding inverse problem is to find pharmacological parameters for given control and drugged AP. Some relevant examples for the latter have been presented by [Bottino et al., 2006 [↗](#)] who estimated pharmacological parameters from APs of canine Purkinje fibers and by Tveito *et al.* who estimated pharmacological parameters from AP biomarkers measured in human induced pluripotent stem cell-derived cardiomyocytes [Tveito et al., 2018 [↗](#)] and several animal ventricular cardiomyocytes [Tveito et al., 2020 [↗](#)]. Furthermore, [Jæger et al., 2021 [↗](#)] identified the optimal polypharmacological treatment for recovering APs of mutant ventricular cardiomyocytes based on biomarkers of simulated wild type and mutant APs.

When the models are interfaced with experimental data, attention should be paid to the inherent uncertainty in the data that results from beat-to-beat variability (intrinsic variability), cell-to-cell variability (extrinsic variability), and measurement errors (observational uncertainty) [Mirams et al., 2016]. Uncertainty propagates through the given problem from APs to estimated parameters or from parameters to predicted APs and must be properly quantified to draw reliable conclusions from the results. Multiple methodologies exist for this purpose [Oakley and O'Hagan, 2004, Mirams et al., 2016, Sher et al., 2022] but usually require many simulations, which even for ordinary differential equation (ODE)-based models of cardiomyocyte EP can become a significant computational burden when considering that each simulation includes a substantial number of beats to reach the model's limit cycle. To overcome this problem, surrogate models have emerged which approximate (emulate) chosen outputs for given inputs multiple orders of magnitude faster. In line with uncertainty quantification literature, the cardiomyocyte EP model is from now on termed the “*simulator*“, whereas the surrogate model is termed the “*emulator*“. Earlier work has reported on an emulator based on linear interpolation of a multi-dimensional lookup table [Mirams et al., 2014] and more recently, Gaussian process (GP) emulators have become popular. Their key advantage is that in- and outputs are modeled as random distributions which allows for rapid sampling of the posterior distributions [Chang et al., 2015, Johnstone et al., 2016, Coveney and Clayton, 2018, Ghosh et al., 2018, Rasmussen and Williams, 2019, Coveney et al., 2021] and while outputs of recently published GP emulators were relevant biomarkers of the AP [Chang et al., 2015, Johnstone et al., 2016, Coveney and Clayton, 2018, Ghosh et al., 2018, Coveney et al., 2021], the emulation of the entire AP can also be realized, e.g. through dimensionality reduction techniques such as the principle component analysis or regressing state-transition models [Mohammadi et al., 2019]. However, GP emulators are not well suited to capture discontinuities of the response surface and may thus fail to capture AP abnormalities, which is a particular drawback for pharmacological studies where bifurcations in behavior such as early afterdepolarizations (EAD) can occur [Ghosh et al., 2018]. To address this, [Ghosh et al., 2018] presented a two-step approach for the emulation of the AP duration at 90% repolarization that first sets up a GP for the location of discontinuities and then fits separate GP emulators for the output of interest either side of these boundaries. In contrast, it has been proven that neural networks (NN) can approximate even discontinuous functions with arbitrary precision in theory [Hornik et al., 1989], while recent works using NNs show empirically promising results for modeling partial differential equations containing discontinuities [Jagtap et al., 2020]. These features render NN emulators suitable emulation candidates and while [Lei and Mirams, 2021] have recently investigated NN emulation of hERG channel kinetics, [Jeong et al., 2023] proposed a neural network using AP shapes as input for the prediction of a drug's proarrhythmic risk. However, to the best of our knowledge, NN emulators have not yet been used as surrogate for cardiomyocyte EP models.

The present study introduces NN emulation of the human ventricular cardiomyocyte AP and investigates the applicability in pharmacological studies. To this end, a NN emulator was developed based on data generated using a state-of-the-art simulator [Tomek et al., 2019, Tomek et al., 2020] and the evaluation was done for forward and inverse problems on synthetic and experimental data.

2 Methods and material

The methodology of this study including the development of the emulator and the evaluation is outlined in **Figure 1**.

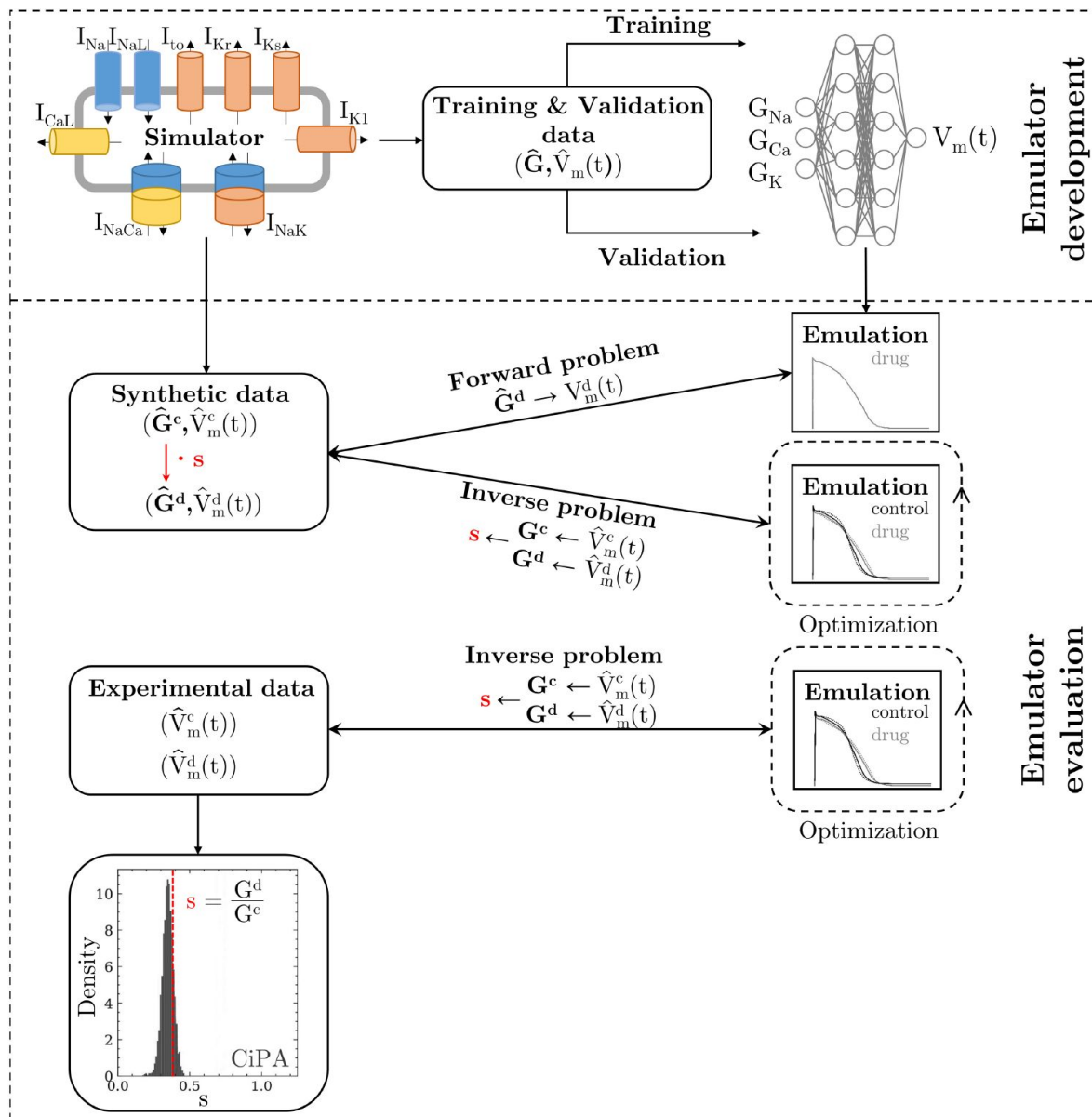


Figure 1

Methodology of this study including the emulator development and the evaluation. The simulator is visualized by a schematic human ventricular cardiomyocyte that includes all currents considered for the emulator training. Inputs of the emulator (see **Figure 2**) are the corresponding maximum conductances ($\mathbf{G}$) but for the sake of illustration, only three representatives are shown. Output is the AP ($V_m(t)$). Training and validation data (maximum conductances $\hat{\mathbf{G}}$ and APs $\hat{V}_m(t)$) were generated by the simulator. The evaluation was performed for forward and inverse problems and to this end, the pharmacological parameter s was introduced. This describes the interaction between the drug and a given target and was defined as scaling factor of the respective maximum conductance in control conditions ($s < 1$: block, $s = 1$: no effect, $s > 1$: enhancement). Synthetic data (control maximum conductances and drugged maximum conductances obtained through scaling, and control and drugged APs) and experimental data (control and drugged APs) were used for the evaluation. The forward problem was only solved for synthetic data, whereas the inverse problem was solved for both synthetic and experimental data. When experimental data were used, estimated pharmacological parameters were compared to distributions derived from data published within the CiPA initiative [Li et al., 2017, Chang et al., 2017].

2.1 Simulator

The simulator of [Tomek et al., 2020] (ToR-ORd-dynCl simulator) was used. This is available in *CARPentry* [Vigmond et al., 2008] and was implemented based on the published *CellML* file for the endocardial subtype¹. Simulations were performed in *CAR-Entry* with the single cell tool *bench*. To compute the gating variables, the Rush-Larsen Method [Rush and Larsen, 1978] was employed, which uses an analytical solution assuming fixed voltage over a small timestep, and the remaining variables were computed by the Forward Euler method. To ensure low computational cost, we found the maximum solver and sampling time steps that still produce accurate results as follows. Various solver and sampling time steps were applied to generate APs and the biomarkers used in this study (see Appendix A) were computed and compared with those that correspond to the minimum time steps (solver: 0.005 ms; sampling: 0.01 ms). The selection criterion was a relative difference below 5% for all AP biomarkers and the chosen time steps (solver: 0.01 ms; sampling: 0.05 ms) led to a maximum relative difference of 2%. APs were stimulated at a pacing cycle length of 1000 ms using 1 ms long rectangular transmembrane current density pulses of $\frac{\mu A}{cm^2}$ [Tomek et al., 2019]. To approach the simulator's limit cycle, a series of 1000 stimuli were applied for each new parameter set starting from the initial states specified in the *CellML* file (when 1000 additional stimuli were applied, the maximum intracellular $[Ca^{2+}]$, $[Cl^-]$, and $[Na^+]$ changed by 1.5%, 0.15%, and 1.7%, respectively). Of note, the simulations can also be performed using the open-source software *openCARP* [Plank et al., 2021].

2.2 Emulator

2.2.1 Architecture

The emulator takes maximum conductances of channels, pumps, and exchangers as inputs and computes the corresponding AP ($V_m(t)$) after the last stimulus as output. It was trained to represent human ventricular cardiomyocytes under control and drugged conditions and the inputs were selected based on two assumptions: 1) The kinetics of channels are preserved, while the number of channels, pumps, and exchangers vary due to different expression levels [Syed et al., 2005, Groenendaal et al., 2015, Krogh-Madsen et al., 2016]. These numbers are captured in the simulator by the maximum conductance parameters (or permeability parameters but maximum conductance is used as general term here for the sake of simplicity) that determine the respective current densities; 2) Channels, pumps, and exchangers are potential (intended and unintended) drug targets and the interaction between drugs and their targets can be described by a scaling of the related maximum conductances [Brennan et al., 2009]. The corresponding scaling factors are pharmacological parameters. These assumptions allowed us to focus on maximum conductances and we considered those as inputs which either the AP is sensitive to (G_{Na} , G_{NaL} , P_{Ca} , G_{to} , G_{Kr} , G_{K1} , G_{NCX} , P_{NaK}) or which are related to common drug targets (G_{Na} , G_{NaL} , P_{Ca} , G_{to} , G_{Kr} , G_{KS} , G_{K1} ; [Crumb et al., 2016]) leading to the following selection: G_{Na} , G_{NaL} , P_{Ca} , G_{to} , G_{Kr} , G_{KS} , G_{K1} , G_{NCX} , and P_{NaK} . AP sensitivity was quantified using a global sensitivity analysis (GSA; see Appendix B) and the inclusion threshold was a total-effect Sobol' sensitivity index (ST) above 0.1 with respect to any of the considered biomarkers (see Appendix A).

Several emulator architectures were tried on the training and validation data sets and the final choice was hand-picked as a good trade-off between high accuracy and low computational cost. We decided to utilize a two-stage emulator architecture: First, the maximum conductances $\mathbf{x}$ – normalized to the range $\mathbf{x}_i \in [-0.5, 0.5]$ – are encoded using a first NN (Θ_1) into a latent representation $\mathcal{V}$. Second, this intermediate representation parameterizes a function $f_{\mathcal{V}}: \mathbb{R} \rightarrow \mathbb{R}$ defined by a second NN (Θ_2) that can be continuously evaluated to receive the emulated AP at time t . To help the second NN in computing the fast depolarization, a simple depolarization term ($\tanh$)

is added to f_{ϑ} . The parameters of this depolarization function are slope (d_1), offset (d_2), and amplitude (d_3), and are created by encoding the parameters through the first network, similar to the latent code. The AP approximated by the emulator is thus defined by

$$V_m(t) := f_{\vartheta}(t) + \tanh(d_1^2(t - d_2)) + d_3^2. \quad (1)$$

A schematic drawing of the emulator architecture is provided in [Figure 2](#). Splitting the network into two parts — one for encoding the parameters into a latent space and a second one for evaluating f_{ϑ} — allowed us to give the emulator enough complexity without markedly increasing the computational cost: in most cases, it is desirable to compute the whole AP in the entire range, e.g. [0, 1000] ms, and not only at a single time step. Creating a single network that computes the mapping from maximum conductances to the transmembrane potential at a single time step (compare [Figure 2](#)) is either orders of magnitudes slower than encoding the parameters into a latent vector (only done once per AP) or would require to reduce the complexity of the network, which led to inaccurate emulations in the validation. The additional depolarization term was introduced to address the difficulty of fitting the depolarization phase during training and decreased the required training time substantially. Note that the mapping from maximum conductances $\mathbf{x}$ to depolarization parameters $\{d_1, d_2, d_3\}$ is also learned through Θ_1 .

The exact architecture employed – chosen by a cross-validation approach (see [Section 2.2.3](#)) – comprised a first network (Θ_1) of four fully connected layers of 256 neurons, each to encode the parameters into the latent vector $\vartheta \in \mathbb{R}^{256}$. This first network additionally generates the parameterization for the depolarization model $\{d_1, d_2, d_3\}$. The second network, computing the APs from the latent representation (Θ_2), consisted of four fully connected layers of 64 neurons each. Exponential linear unit (ELU) activation functions [[Clevert et al., 2016](#)] were used for all layers, except for the final non-linear layer, which was modeled using a tanh activation function followed by a $[-1, 1]^{64} \rightarrow \mathbb{R}$ linear layer.

2.2.2 Training

The supervised training data set X contained pairs of maximum conductance samples $\mathbf{x}$ and the corresponding AP $\hat{V}_m(t)$ that was obtained from the simulator. Sobol' sequences were used to generate 40,000 maximum conductance samples, containing 20,000 maximum conductance samples between 0% and 200% of the original values and 20,000 maximum conductance samples between 50% and 150% of the original values. The first covers a range that was considered plausible in terms of physiology and pathology [[Britton et al., 2017](#), [Tomek et al., 2019](#)], and in terms of pharmacology (where full block is plausible). The latter covers a range that was considered particularly relevant in line with experimental calibration results presented in [[Tomek et al., 2019](#)] and patch clamp measurements of channels that were exposed to 30 clinical drugs (blocks in up to the four-fold of the maximum free therapeutic concentration were analyzed in agreement with the CiPA paradigm [[Li et al., 2019](#)] [[Crumb et al., 2016](#)]). The SALib-Sensitivity Analysis Library [[Herman and Usher, 2017](#)] was used in the entire study for the generation of samples based on Sobol' sequences. For each maximum conductance sample, simulations were performed to obtain the corresponding 40,000 APs. APs with a transmembrane potential difference of more than 10% of the amplitude between $t = 0$ and 1000 ms were excluded. This, however, applied to only 116 APs.

Starting from the original APs, the data were first extended by 10 ms from $t \in [0, 1000]$ ms to $t \in [-10, 1000]$ ms to enable some extrapolation of V_m and hence a better alignment of the depolarization; for $t \in [-10, 0]$ ms the initial resting membrane potential $V_m(0)$ was held constant. Then, the data were non-uniformly resampled to 4 kHz for $t \in [-20, -5]$ ms (resting phase) and 10 kHz for $t \in [-5, 20]$ ms (depolarization phase). The repolarization phase ($t \in [20, 1000]$ ms) was again resampled to 4 kHz.

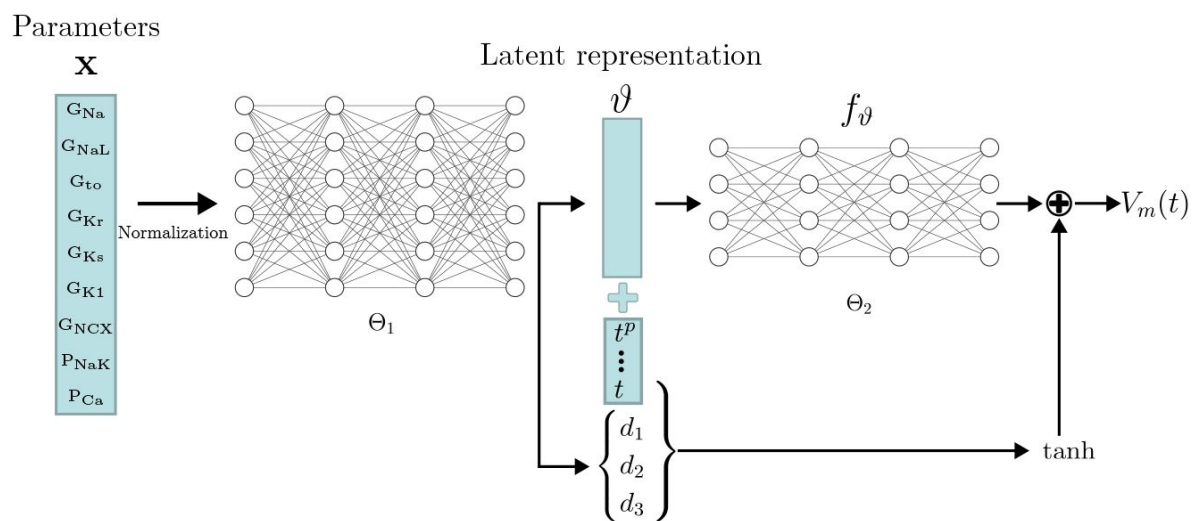


Figure 2

Conceptual architecture of the neural network emulator. The maximum conductances $\mathbf{x}$ are encoded into depolarization parameters d_i and a latent space representation ϑ that uniquely defines the time series functional $f_\vartheta: \mathbb{R} \rightarrow \mathbb{R}$. The time is normalized and encoded in polynomials up to degree 8 (t^p for $p \in \{1, 2, \dots, 8\}$), before being appended to the latent code ϑ . f_ϑ is then used in conjunction with the depolarization helper $\tanh$ to approximate the AP $V_m(t)$.

From the initial training data set, 20% were randomly excluded to be used for validation leaving 31908 pairs of maximum conductances and corresponding APs for training. **Figure 3** shows processed APs that were used for training and validation. For better visualization, only $\approx 25\%$ are included but this already illustrates the large variety of APs spanning from normal to abnormal APs.

Although different regularization schemes such as variational losses (e.g. $\frac{\partial V_m}{\partial t}$) were tried, the wealth of training data allowed us to define the training loss purely in terms of mean-squared-error (MSE)

$$\mathcal{L}(\Theta_1, \Theta_2) = \frac{1}{2|\mathbb{T}|} \sum_{(\mathbf{x}, \hat{V}_m) \in \bar{X} \subset X} \sum_{t \in \mathbb{T}} \left(f_{\vartheta_{\Theta_1}(\mathbf{x}), \Theta_2}(t) - \hat{V}_m(t) \right)^2 dt, \quad (2)$$

where $f_{\vartheta_{\Theta_1}(\mathbf{x}), \Theta_2}$ describes the output of the emulator using the current NN weights Θ_1/Θ_2 , $\bar{X}$ refers to the current training batch and X is the training data set containing both target APs $\hat{V}_m$ and corresponding maximum conductances $\mathbf{x}$. For the training, increasing batch sizes ($|\bar{X}|$ 1250 to 1800) were used, both in terms of the entire AP and considering only subsets over time $T \subset \mathbb{T}$ from $\frac{1}{16}|\mathbb{T}|$ to $|\mathbb{T}|$, resulting in a reduction of training time needed. The neural network was trained for a total number of 5000 epochs using the first-order gradient-based algorithm ADAM [Kingma and Ba, 2017]. The training time was approximately 4 h on a Nvidia A100 (Nvidia Corporation, USA). The simulation of 1000 APs (see Section 2.1) sampled at a resolution of 20 kHz took 293 s on one core of a AMD Ryzen Threadripper 2990WX (clock rate: 3.0 GHz). The emulation of an AP (that approached the limit cycle after 1000 stimuli) sampled at a resolution of 20 kHz for $t \in [-10, 1000]$ ms took 18.7 ms on a AMD Ryzen 7 3800X (clock rate: 4.5 GHz) and 1.59 ms on a Nvidia GeForce GTX 1660 Super GPU, including synchronization and data copy overhead. This is a speed-up of $1.57 \cdot 10^4$ and $1.84 \cdot 10^5$, respectively.

2.2.3 Validation

A cross-validation approach was used to quantify and compare the performance of various emulator architectures. The validation was based on 20% of the initial training data set (7976 pairs of maximum conductances and corresponding APs, see Section 2.2.2). For each pair, the emulated AP V_m was compared against the simulated AP $\hat{V}_m$, given the same maximum conductances. The mismatch was quantified by the root-mean-squared error (RMSE) defined as

$$\text{RMSE}(V_m, \hat{V}_m) := \sqrt{\frac{1}{|\mathbb{T}|} \int_T \left(V_m(t) - \hat{V}_m(t) \right)^2 dt}.$$

The mismatch was also quantified in terms of AP biomarkers $\mathbf{b} \in \mathbb{R}^N$ (see Appendix A) and normalized maximum conductances $\mathbf{x}$ (see Section 2.2.4) in which case the RMSE was defined as

$$\text{RMSE}(\mathbf{b}, \tilde{\mathbf{b}}) := \sqrt{\frac{1}{N} \|\mathbf{b} - \tilde{\mathbf{b}}\|_2^2} \quad \text{and} \quad \text{RMSE}(\mathbf{x}, \tilde{\mathbf{x}}) := \sqrt{\frac{1}{N} \|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2},$$

for N samples.

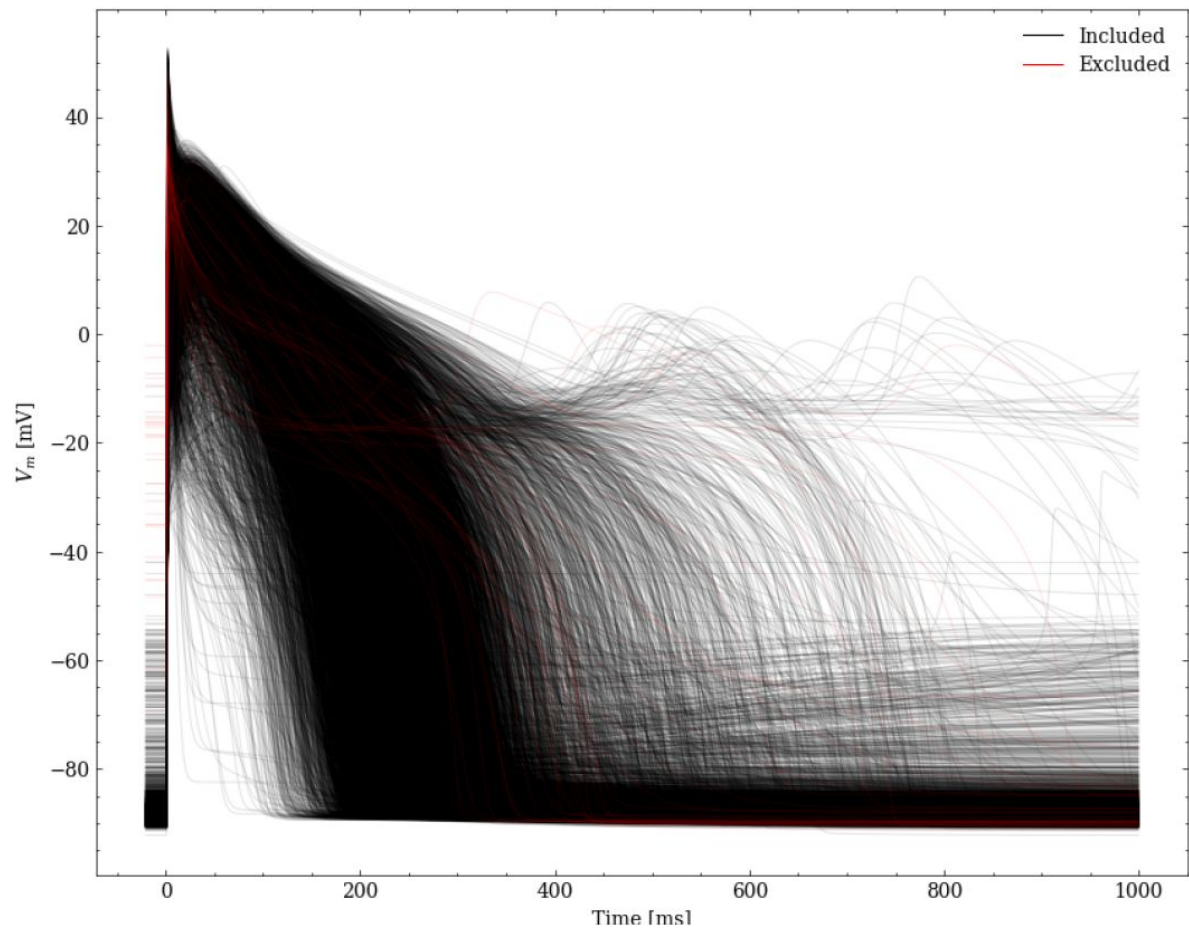


Figure 3

Processed APs used for training and validation. For better visualization, only 4000 of the set are shown.

2.2.4 Time series fitting and estimation of maximum conductances and pharmacological parameters

Time series fitting is the basis for solving the inverse problem. To fit a given AP $\hat{V}_m$, defined on a subset of the trained domain $\hat{\mathbb{T}} \subset \mathbb{T}$, the first step was to choose a trial set of maximum conductances $\mathbf{x}_0$. Then, for the given trial set, the corresponding AP was emulated and the trial set of maximum conductances was iteratively updated by solving the following minimization problem:

$$\min_{\mathbf{x}, t_0} \frac{1}{2|\hat{\mathbb{T}}|} \sum_{t \in \hat{\mathbb{T}}} \left(V_m(\mathbf{x}, t - t_0) - \hat{V}_m(t) \right)^2 dt + \frac{\lambda_{\mathbf{x}_0}}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \delta_{[-0.5, 0.5]}(\mathbf{x}), \quad (3)$$

where $V_m(\mathbf{x}, t) := f_{\vartheta_{\Theta_1}(\mathbf{x}), \Theta_2}(t)$ is a shorthand for the emulator approximation function and t_0 is a temporal offset parameter helping in fitting the exact depolarization timing. Here, $\delta_{[-0.5, 0.5]}(\mathbf{x})$ is the element-wise indicator function on the normalized feasible parameter space $[-0.5, 0.5]$. The minimization was done using ADAM [Kingma and Ba, 2017] combined with a projection on the feasible space.

To estimate maximum conductances for a given control AP, the control AP was fitted using the original maximum conductance values as initial guesses and priors: $\mathbf{x}_0 = \hat{\mathbf{x}} = \mathbf{0}$. To estimate maximum conductances for a given drugged AP, the drugged AP was fitted using the maximum conductances estimated for the respective control AP as initial guesses and priors: $\mathbf{x}_0 = \hat{\mathbf{x}} = \mathbf{x}^c$. The pharmacological parameters (scaling factors of control maximum conductances) were computed as element-wise ratios between the drugged and control maximum conductances $(s)_i := \frac{(G^d)_i}{(G^c)_i}$ but here, G^d and G^c are the non-normalized maximum conductances, where $(G^c)_i > 0$.

Since multiple sets of maximum conductances produced similarly good fits of the given AP, the parameter $\lambda_{\mathbf{x}_0}$ was introduced which minimizes the difference between original and control maximum conductances and between control and drugged maximum conductances, respectively. The value was chosen to be $\lambda_{\mathbf{x}_0} = 10$ with respect to the highest accuracy found for the synthetic data set #1 that was generated for the evaluation (see Section 2.3.1).

2.3 Evaluation

The evaluation was performed for forward and inverse problems in pharmacological studies on synthetic and experimental data.

2.3.1 Data

Synthetic data

Two sets of synthetic data were created using the simulator and each of the sets consisted of control and drug data with pairs of maximum conductances and corresponding APs.

The control data were the same in both sets. They were created using an experimentally calibrated population of 100 synthetic cardiomyocytes [Britton et al., 2013, Muszkiewicz et al., 2016] and to this end, Sobol' sequences were used to generate samples of maximum conductances with values between 50% and 150% of the original values. Maximum conductance samples that produced APs without abnormalities (checked for the last two consecutive APs; see Appendix A), and with seven biomarker values (derived from the last AP; see Appendix A) in agreement with experimental ranges (Table 1) were included in the population. Please note that the experimental ranges were not derived from the data set described in Section 2.3.1.

AP biomarker	Unit	Min	Max
<i>RMP</i>	mV	−95	−80
<i>dVmMax</i>	mVms ^{−1}	100	1000
<i>Peak</i>	mV	10	55
<i>APD₄₀</i>	ms	85	320
<i>APD₅₀</i>	ms	110	350
<i>APD₉₀</i>	ms	180	440
<i>Tri_{90–40}</i>	ms	50	150

Table 1

AP biomarkers and their experimental ranges used to generate the population of synthetic cardiomyocytes. These were adopted from [Passini et al., 2017]. Experimental data were collected at 37 °C in small right ventricular trabeculae and papillary tissue preparations obtained from healthy human hearts during pacing with a cycle length of 1000 ms [Britton et al., 2017].

Data set #1: The motivation for creating data set #1 was to evaluate the emulator on data of normal APs.

Drug data were created using 100 synthetic drugs represented by a set of pharmacological parameters. Each synthetic drug was built to have four different targets, with all channels, pumps, and exchangers related to the emulator inputs considered as potential targets. To this end, 100 samples of four pharmacological parameters, each with values between 0.5 (50% block) and 1.5 (50% enhancement) were randomly generated. The synthetic drugs were applied to the entire synthetic cardiomyocyte population by scaling each of the relevant control maximum conductances with the corresponding pharmacological parameter. The samples that produced APs without abnormalities (checked for the last two consecutive APs; see Appendix A) were included in the data set. No sample was excluded and thus, the data set consisted of 100 control data pairs and 10,000 drug data pairs.

Data set #2: The motivation for creating data set #2 was to test the emulator on data of abnormal APs showing the repolarization abnormality EAD. This is considered a particularly relevant AP abnormality in pharmacological studies because of their role in the genesis of drug-induced ventricular arrhythmia's [Weiss et al., 2010]. Drug data were created using ten synthetic drugs with the hERG channel and the Cav1.2 channel as targets. To this end, ten samples with pharmacological parameters for G_{Kr} and P_{Ca} (Table 2) were generated and the synthetic drugs were applied to the entire synthetic cardiomyocyte population by scaling G_{Kr} and P_{Ca} with the corresponding pharmacological parameter. The samples that produced APs with EADs (see Appendix A) but a transmembrane potential difference of not more than 10% of the amplitude between $t = 0$ and 1000 ms (checked for the last AP) were included. The data set consisted of 100 control data pairs and 122 drug data pairs.

Experimental data

APs were recorded in small right ventricular trabeculae and papillary tissue preparations that were obtained from healthy human hearts [Orvos et al., 2019]. The conventional microelectrode technique was used for recording at a temperature of 37 °C. Stimulation of APs was done at a pacing cycle length of 1000 ms using a pair of platinum electrodes that provided rectangular current pulses of 2 ms duration. To allow the preparations to equilibrate, stimuli were delivered for at least 60 min before the measurements started. Measurements were performed under control conditions and after administration of five channel-blocking drugs at one concentration in multiple preparations. Drugs were cisapride (30 nM), dofetilide (10 nM), sotalol (30 µM), terfenadine (1 µM), and verapamil (300 nM). The last ten consecutive APs of each measurement were analyzed to quantify the beat-to-beat variability. Overall, the beat-to-beat variability was found to be small (standard deviation in all APs below 7 mV before the peak due to time alignment mismatch and below 2 mV after the peak) and thus, the last ten consecutive APs of each measurement were averaged. In most of the preparations, the standard deviation between beats did not vary over time and thus, no temporal correlation of noise was assumed. Averaging also reduced the noise level. The experimental data set contained one pair of averaged control and drugged AP per preparation per drug. Pairs were excluded if the biomarker values (see Appendix A) of the control or the drugged AP were not in the range found in the training data (see Section 2.2.2). This applied to seven pairs and the final data set contained three pairs for cisapride, dofetilide, sotalol, and terfenadine, and one pair for verapamil. All measurements were performed at the University of Szeged, Hungary, and conformed to the principles of the Declaration of Helsinki. Protocols were approved by the National Scientific and Research Ethical Review Boards (4991-0/2010-1018EKU [339/PI/010]).

The raw data were obtained without filtering but some filtering was applied before interfacing the data with the emulator. The APs contained a stimulus artifact between 0 and 1.5 ms that was filtered as follows. For each of the last ten consecutive APs, the transmembrane potential V_m closest to the end of the recorded time series was defined as the resting transmembrane potential

ID	1	2	3	4	5	6	7	8	9	10
G_{Kr}	0.05	0.06	0.07	0.08	0.09	0.10	0.11	0.12	0.13	0.14
P_{Ca}	1.20	1.22	1.24	1.26	1.28	1.30	1.32	1.34	1.36	1.38

Table 2

Pharmacological parameter samples (synthetic drugs) with scaling factors for G_{Kr} and P_{Ca} to generate the drug data of data set #2.

RMP and $V_m(t) = RMP$ was set for $t \in [0, 1.75]$ ms. Then, the APs were resampled at 100 kHz and a low-pass filtering was performed with a second-order butterworth filter (cutoff at 2.5 kHz) to reduce the high-frequency noise of the signal. Finally, the filtered APs were averaged and the averaged AP was again resampled at 1 kHz for $t \in [15, 1000]$ ms (repolarization) and 100 kHz in $t \in [0, 15]$ ms (depolarization). An example comparison of a raw and a filtered averaged AP is given in [Figure 4](#).

2.3.2 Forward problem

The emulator evaluation for the forward problem, i.e. to find the drugged AP for given pharmacological parameters, was only performed on synthetic data since maximum conductances were not available experimentally. The data sets #1 and #2 were used to consider data with normal APs and with abnormal APs exhibiting EADs. Pharmacological parameters are not inputs of the emulator but drugged maximum conductances that were computed as control maximum conductances scaled by the given pharmacological parameters (see [Section 2.3.1](#)). These were used to emulate drugged APs. The RMSE was used to quantify the mismatch between the emulated and the ground truth AP.

2.3.3 Inverse problem

The emulator evaluation for the inverse problem, i.e. to find the pharmacological parameters for given control and drugged APs (through optimization), was performed on both synthetic and experimental data. When using synthetic data, the data set #1 was used including data with normal APs. First, control and drugged maximum conductances were estimated based on control and drugged APs and then, pharmacological parameters were computed as ratios of drugged and control conductances (see [Section 2.2.4](#)). The mismatch between estimated and ground truth maximum conductances were quantified using the error that is defined through

$$\mathbf{e}_n := (\mathbf{x} - \hat{\mathbf{x}}), \quad (4)$$

where $\mathbf{x}$ and $\hat{\mathbf{x}}$ are the normalized estimated and ground truth maximum conductances (see [Section 2.2.1](#)). Similarly, we computed the mismatch between estimated and ground truth scaling factor vectors ($\mathbf{s}$ and $\hat{\mathbf{s}}$ respectively) as

$$\mathbf{e}_s := (\mathbf{s} - \hat{\mathbf{s}}). \quad (5)$$

When using experimental data, maximum conductances and pharmacological parameters were estimated in the same way but due to a lack of experimental maximum conductances, the mismatch between estimated and ground truth values could not be quantified. Instead, the estimated pharmacological parameters were compared with distributions computed from data published within the CiPA initiative [[Li et al., 2017](#), [Chang et al., 2017](#)] (CiPA distributions). The data set ² includes 2000 IC_{50} values and Hill coefficients for each drug and for up to seven targets (I_{Na} , I_{NaL} , I_{CaL} , I_{to} , I_{Kr} , I_{Ks} , I_{K1}). The pore-block model [[Brennan et al., 2009](#)] was used to obtain the corresponding scaling factors.

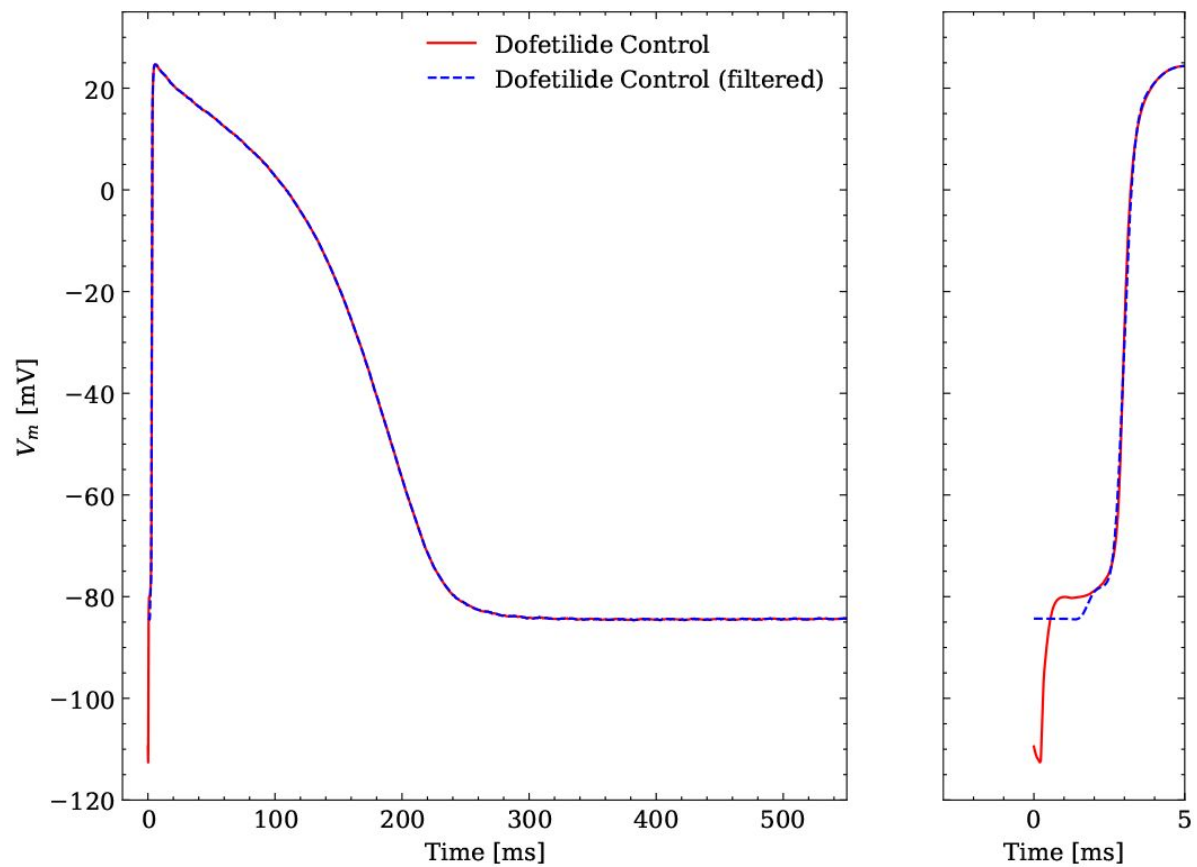


Figure 4

Comparison of an averaged raw and an averaged filtered experimental AP. One dofetilide control AP is shown as example.

3 Results

3.1 Evaluation

3.1.1 Forward problem

The emulator evaluation for the forward problem was only done on synthetic data and both data sets #1 and #2 (see [Section 2.3.1](#)) were used to analyze the solution accuracy for normal and abnormal APs exhibiting EADs.

The data set #1 was used first and [Figure 5](#) illustrates the distribution of RMSEs between emulated and ground truth drugged APs. The average RMSE over all APs was 0.47 mV and only for a few APs the RMSE was >1 mV with 1.5 mV being the maximum. Largest mismatches were located in the phases 0 and 3 of the AP. While the mismatches in phase 3 were simply a result of imperfect emulation, the mismatches in phase 0 were a result of the difficulty in matching the depolarization time exactly.

[Figure 6](#) shows the distribution of biomarker mismatches between emulated and ground truth drugged APs. The low RMSEs between the APs translated into low RMSEs between the AP biomarkers.

The data set #2 was used second and Appendix C shows all emulated APs. In total, 72% of the APs exhibited EADs (classified based on the criterion outlined in Appendix A) in agreement with the ground truth drugged APs, and the substantial majority of the remaining APs showcase a notable proximity to the threshold of manifesting EADs. [Figure 7](#) illustrates the distribution of RMSEs between emulated and ground truth drugged APs. The average RMSE over all APs was 13.6 mV with 43.2 mV being the maximum. Largest mismatches were located in phase 3 of the AP, in particular in emulated APs that did not fully repolarize.

3.1.2 Inverse problem based on synthetic data

The emulator evaluation for the inverse problem was first done using synthetic data (data set #1, see [Section 2.3.1](#)). Control and drugged APs could be fitted with an average RMSE of 0.8 mV. Largest mismatches were located in phase 0 and 3 of the AP for the reasons given above (see [Section 3.1.1](#)). [Figure 8](#) shows the distribution of the errors between the estimated and the ground truth maximum conductances and pharmacological parameters. For both the maximum conductances ($\text{RMSE} \leq 0.18$) and the related pharmacological parameters ($\text{RMSE} \leq 0.21$) the errors were closely distributed around zero. However, the RMSEs increased from the control maximum conductance over the drugged maximum conductance to the pharmacological parameters and there were distinctive differences among maximum conductances and related pharmacological parameters with the smallest for G_{K_r} and the largest for G_{K_s} .

3.1.3 Inverse problem based on experimental data

The emulator evaluation for the inverse problem was then done on experimental data ([Section 2.3.1](#)).

[Figure 9](#) shows the fitted and the ground truth APs for all drugs. Control and drugged APs could be fitted with average RMSEs shown in [Table 3](#). The largest mismatch was located in phase 0 for most APs, which was the result from an imperfect matching of the exact depolarization timing.

[Figure 10](#) compares the estimated pharmacological parameters and the CiPA distributions. Pharmacological parameters that fell into the range spanned by $\mu \pm (0.15 + \sigma)$ of the CiPA distribution, where μ , σ are the distribution's mean and standard deviation respectively, were

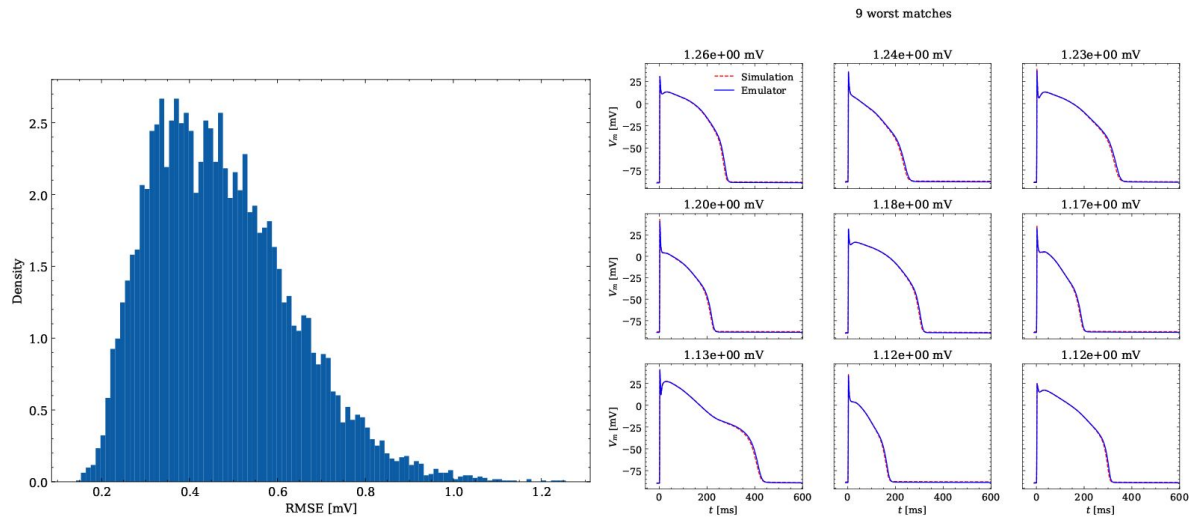


Figure 5

Analysis of solution accuracy of the forward problem on synthetic data including normal APs (drug data of data set #1). Left: histogram of RMSEs for the APs, right: APs with the largest RMSEs. The RMSE is given above each subplot.

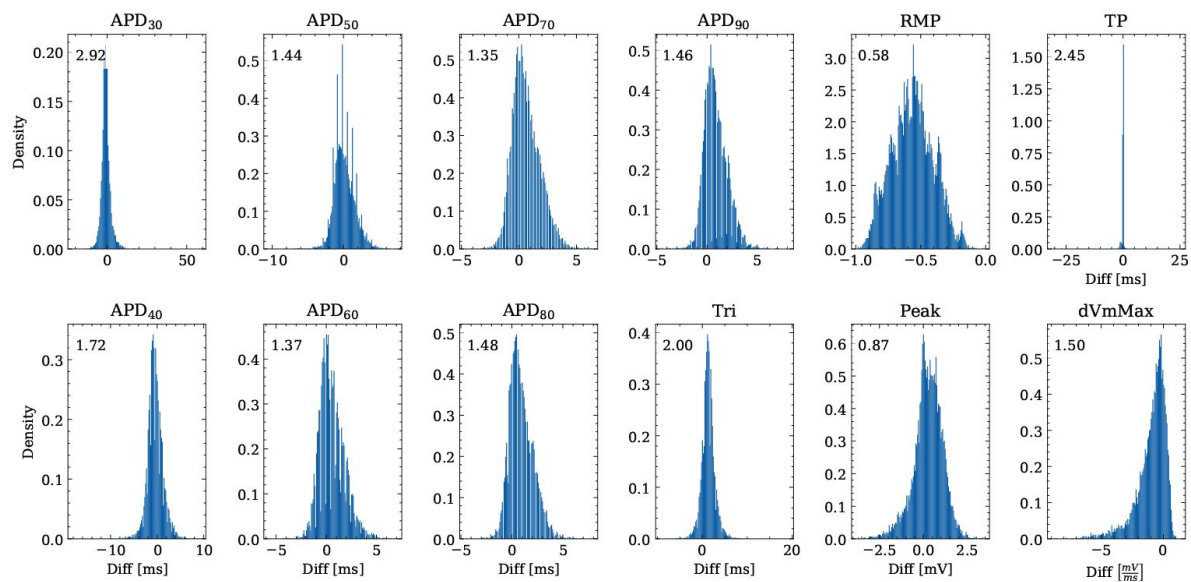


Figure 6

Analysis of solution accuracy of the forward problem on synthetic data including normal APs (data set #1) with respect to AP biomarkers. Histograms of mismatches for each biomarker are shown and the RMSE is given in the upper left corner. Note that potential staircasing artifacts in the APDs are related to the temporal discretization of the APs generated from the simulator.

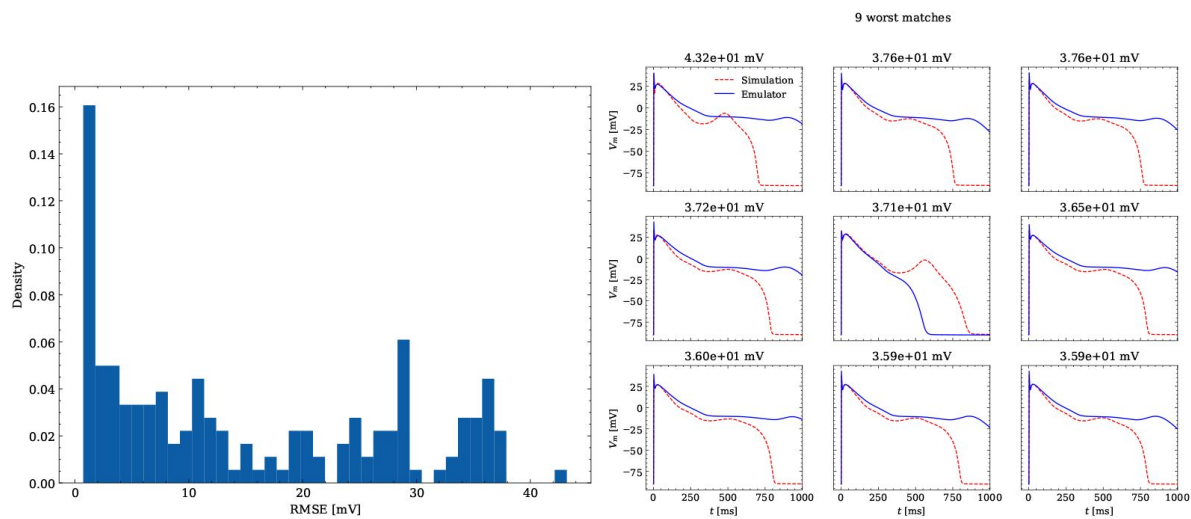


Figure 7

Analysis of solution accuracy of the forward problem on synthetic data including abnormal APs exhibiting EADs (drug data of data set #2). Left: histogram of RMSEs for the APs, right: APs with the largest RMSEs. 123 of the 170 emulated traces exhibit the expected EADs (based on the criterion outlined in Appendix A). The RMSE is given above each subplot. All emulated APs are shown in Appendix C

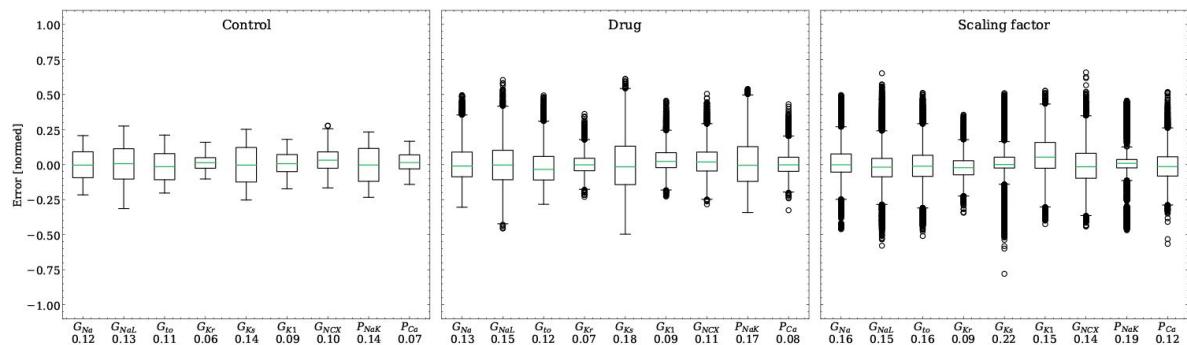


Figure 8

Analysis of solution accuracy of the inverse problem on synthetic data (data set #1). Left: boxplot of errors between normalized estimated and ground truth control maximum conductances, middle: boxplot of errors between normalized estimated and ground truth drugged maximum conductances, right: boxplot of errors between estimated and ground truth pharmacological parameters. Error definitions are given in (4) and (5). The RMSE over all data is given below each parameter.

Drug	RMSE Control	RMSE Drug
Cisapride	1.53	2.00
Dofetilide	2.05	1.73
Sotalol	1.40	2.51
Terfenadine	1.22	1.08
Verapamil	1.93	2.21

Table 3

Average RMSE over control and drugged APs measured in all preparations per drug. All values in mV.

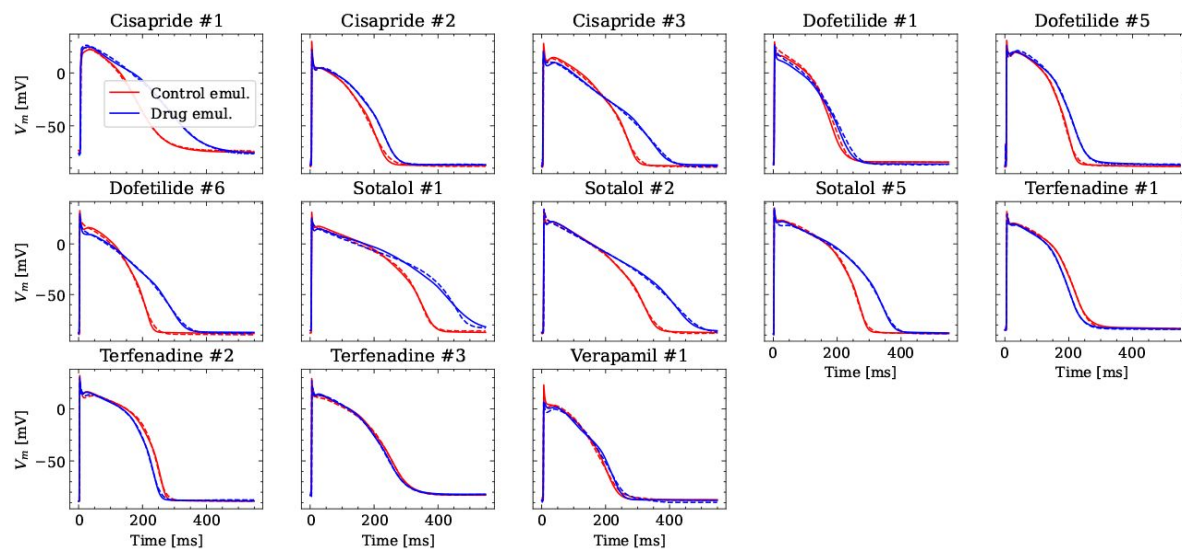


Figure 9

Analysis of fit quality of the inverse problem on experimental data. Comparison of the fitted APs (solid lines) and the experimental APs (dashed lines) at control (red) and after drug administration (blue) for all drugs.

classified as successfully estimated while the others were classified as unsuccessfully estimated. In total, 50% of the pharmacological parameters could be estimated successfully and while all pharmacological parameters related to G_{Ks} could be successfully estimated, unsuccessfully estimated parameters were found across all maximum conductances, in particular related to G_{K1} for which all pharmacological parameters could not be successfully estimated (**Table 4** [↗](#)).

4 Discussion

NN emulation of the human ventricular cardiomyocyte AP was introduced and the applicability in pharmacological studies was investigated.

4.1 Evaluation

The presented NN emulator enables a massive speed-up compared to regular simulations ($> 10^4$) and the evaluation for the forward problem on synthetic data showed also highly accurate AP emulations. High accuracy was found in normal APs (average RMSE was 0.47 mV; **Figure 5** [↗](#)) and mostly also in abnormal APs exhibiting EADs (of the emulated APs, 72% exhibited alignment with the abnormality, and the substantial majority of the remaining APs demonstrated pronounced proximity, while the average RMSE was 13.6 mV; **Figure 7** [↗](#)). Increasing the amount of training data within the relevant range could lead to further enhancements in accuracy for abnormal APs. Nevertheless, this observation demonstrates that the emulator is also capable of accounting for discontinuities of the response surface. This is particularly useful in pharmacological studies and a key advantage over existing emulation approaches [Chang et al., 2015 [↗](#), Johnstone et al., 2016 [↗](#), Coveney and Clayton, 2018 [↗](#), Ghosh et al., 2018 [↗](#), Rasmussen and Williams, 2019 [↗](#), Coveney et al., 2021 [↗](#)].

The emulator was further evaluated for the inverse problem on synthetic and also on experimental data. Maximum conductances and related pharmacological parameters could be widely estimated with high accuracy on synthetic data ($RMSE \leq 0.18$ and ≤ 0.21 for all maximum conductances and pharmacological parameters, respectively; **Figure 8** [↗](#)).

The RMSEs increased from the control maximum conductance over the drugged maximum conductance to the pharmacological parameters which may be because the estimation of drugged maximum conductances depends on the control maximum conductances and the computation of pharmacological parameters depends on both control and and drugged maximum conductances (see *Section 2.2.4* [↗](#)) allowing that errors can propagate and amplify. Distinctive differences were observed among the maximum conductances and related pharmacological parameters and the largest RMSEs were found for G_{Ks} throughout. This can be attributed to various degrees of parameter identifiability [Sarkar and Sobie, 2010 [↗](#), Zaniboni et al., 2010 [↗](#), Groenendaal et al., 2015 [↗](#), Jæger et al., 2019 [↗](#), Whittaker et al., 2020 [↗](#)] and the results agree with the GSA that indicates almost non-identifiability of G_{Ks} (Appendix B).

Larger inaccuracies were found in the inverse problem solutions on experimental data (**Figure 10** [↗](#), **Table 3** [↗](#)). The first reason may be low parameter identifiability and we want to highlight inaccuracies in estimating the pharmacological parameters related to G_{Kr} , P_{Ca} , and G_{NaL} when the hERG channel was blocked in parallel to the Cav1.2 channel (verapamil) or in parallel to both the Cav1.2 and the Nav1.5-late channel (terfenadine). The hERG channel block (prolongation of the AP), and the Cav1.2 and Nav1.5-late channel block (shortening of the AP) are known to have opposite effects on the AP [Orvos et al., 2019 [↗](#)]. At the given drug concentrations, these effects were apparently counterbalancing, which resulted in negligible changes of the AP (**Figure 9** [↗](#)). This situation made the estimation of pharmacological parameters very challenging and led to particularly large inaccuracies for terfenadine. The accurate estimations of the pharmacological parameters related to G_{Ks} are surprising at first in light of the almost non-identifiability. This was

	G_{Na}	G_{NaL}	G_{to}	G_{Kr}	G_{Ks}	G_{K1}	P_{Ca}	Total
Successful	6	6	8	3	9	0	5	37
Unsuccessful	3	1	5	10	0	10	8	37
Ratio	0.67	0.86	0.62	0.23	1.00	0.00	0.38	0.50

Table 4

Pharmacological parameters related to maximum conductances that were considered successfully or unsuccessfully estimated across all preparations and drugs

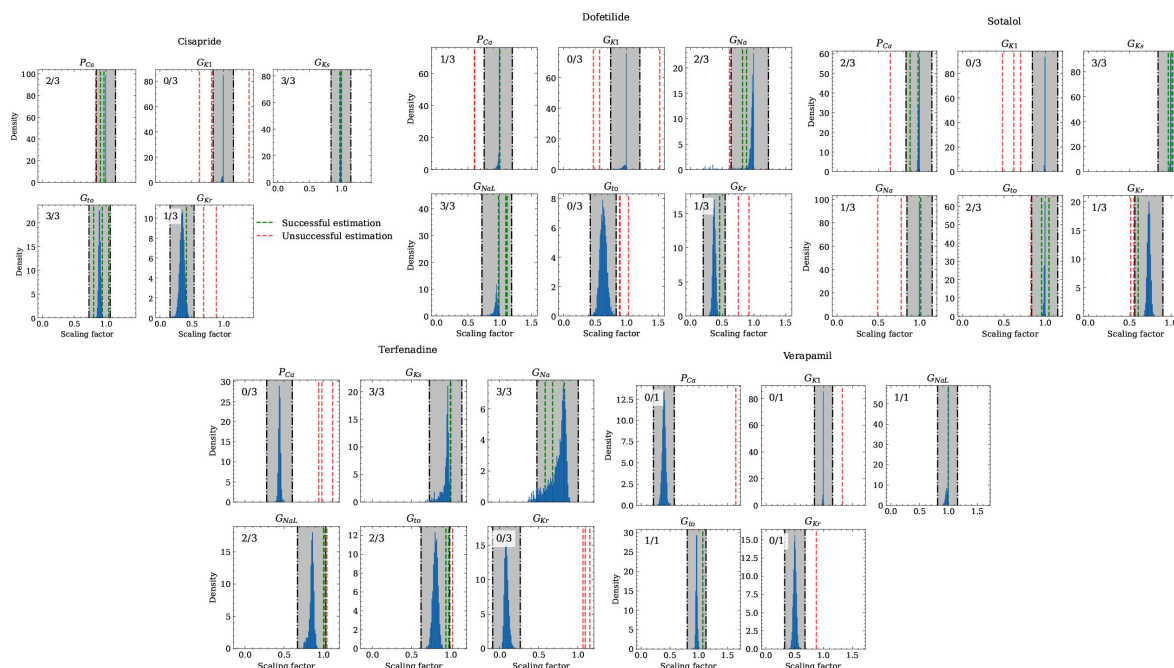


Figure 10

Analysis of solution accuracy of the inverse problem using experimental data. The histograms compare the estimated pharmacological parameters (dashed vertical lines) from data of multiple preparations with the CiPA distributions (blue; see Section 2.3.3). The black dash dotted vertical lines are the borders of the range (grey) that was used to determine if the estimation of the given pharmacological parameter was successful. The range is spanned up by $\mu \pm (0.15 + \sigma)$ of the CiPA distribution. Successfully estimated parameters are shown as green lines and unsuccessfully estimated parameters are shown as red lines. The number in the upper left corner indicates for how many preparations the parameters could be successfully estimated in relation to the total number of preparations for the given drug.

due to the combination of two factors: 1) different from synthetic data, the drugs at the given concentrations did not affect the corresponding KCNQ1-MinK channel and 2) the difference between control and drugged maximum conductance was weakly enforced to be minimal (see [Section 2.2.4](#)) which leads to almost no difference in non-identifiable parameters and hence, to a pharmacological parameter of one. The accuracy will likely be much lower in drugs that affect the KCNQ1-MinK channel.

The second and probably main reason for the inaccuracies may be the fact that the data were collected in small tissue preparations, whereas the emulator was trained on data generated by a simulator that represents single cardiomyocytes. APs in small tissue preparations are slightly different from those in single cardiomyocytes. Differences can arise from electrotonic coupling and the mixture of cells including fibroblasts that are able to modify the EP [\[Kohl and Gourdie, 2014\]](#), [Mayer et al., 2017\]](#), [Hall et al., 2021\]](#). This can hamper the fitting of the APs and consequently, the estimation of the maximum conductances and pharmacological parameters.

4.2 Emulator

In addition to the aforementioned strengths of the presented NN emulator, some further valuable features are worth mentioning that arise from the continuous nature of the emulator. First, the AP can be emulated and fitted at any desired resolution. Second, timing offsets, e.g. between stimuli in the data to be fitted and the training data, can be accounted for using t_0 in (3) without retraining. Last but not least, the transmembrane potential gradient $\frac{\partial V_m}{\partial t}$, recently highlighted in terms of proarrhythmic potential prediction [\[Jeong et al., 2022\]](#), can be continuously derived and is not dependent on the temporal discretization.

4.3 Limitations and future work

Some limitations have to be considered. First, the emulator has only maximum conductances as inputs. Although these explain much of the AP variability seen between cardiomyocytes [\[Britton et al., 2013\]](#), [Muszkiewicz et al., 2016\]](#), the inclusion of parameters related to the channel kinetics might enable a more detailed consideration of drug effects in pharmacological studies. Channel kinetics determine the contribution of the corresponding current to the AP generation in different phases and can thus also modulate drug effects. However, the expansion of the input space might be unsuitable for solving the inverse problem when only AP data are used. Second, the interaction between drugs and their targets is solely captured through scaling of the related maximum conductance at control, which is mostly adequate but in fact an oversimplification [\[Brennan et al., 2009\]](#). The interaction can be dependent on time, voltage, and channel state, which requires the use of Markov models with multiple more parameters [\[Brennan et al., 2009\]](#), [Li et al., 2017\]](#). Again, this expands the input space and might be unsuitable for solving the inverse problem when only AP data are used. Moreover, drugs that are applied over a longer period of time can also cause modifications of the maximum conductances through changes in gene expression [\[Shim et al., 2023\]](#). This requires attention to avoid misinterpretations of found blocking or enhancement effects, e.g. by estimating the control maximum conductances again after a washout procedure. Third, the inverse problem was only solved for AP data obtained from one single stimulation protocol. [\[Johnstone et al., 2016\]](#) have shown that the usage of AP data obtained from various stimulation protocols can improve the parameter identifiability and thus, the accuracy of parameter estimates. To be able to use those data in the presented approach, the pacing cycle length must be included as additional input in the emulator and the emulator may be trained on more than the last AP of the pacing series. This would also allow to capture alternans. Last but not least, the number of drugs and concentrations considered in the inverse problem on experimental data poses a limitation. The ultimate goal is to have a tool that provides highly accurate solutions for drugs with different targets and concentrations. To this end, analyses must be extended by data obtained from a series of available and well characterized drugs. The data should be collected in single cardiomyocytes in order to minimize the discussed inaccuracies that stem from the use of

tissue preparation data. This should be addressed in future work. Of note, the presented approach can also be straightforwardly applied to other transients, e.g. of the intracellular $[Ca^{2+}]$ or of the sarcomere length.

4.4 Conclusion

This paper introduced NN emulation of the human ventricular cardiomyocyte AP and tested its applicability in pharmacological studies. The computational cost of the NN emulator was compared to that of the simulator, revealing a massive speed-up of more than 10^4 . The accuracy of solving the forward problem on synthetic data was found to be high for normal APs and this holds mostly true for abnormal APs exhibiting EADs. This advantage distinguishes our novel approach from existing emulation methods. While larger inaccuracies were observed when utilizing experimental data – a limitation thoroughly discussed and particularly inherent to the fact that small tissue preparations were studied while the emulator was trained on single cardiomyocyte data – the accuracy of solving the inverse problem on synthetic data remained high. Collectively, these findings underscore the potential of NN emulators in improving the efficiency of future quantitative systems pharmacology studies.

5 Code availability

The trained emulator is available as a python package under the link https://github.com/thomgrand/cardiomyocyte_emulator³.

6 Additional information

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6.2 Author contributions

Thomas Grandits: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing

Christoph M. Augustin: Funding acquisition, Writing – review & editing

Gundolf Haase: Funding acquisition, Resources, Writing – review & editing

Norbert Jost: Funding acquisition, Resources, Writing – review & editing

Gary Mirams: Funding acquisition, Methodology, Writing – review & editing

Steven Niederer: Funding acquisition, Writing – review & editing

Gernot Plank: Funding acquisition, Resources, Writing – review & editing

László Virág: Resources, Writing – review & editing

András Varró: Funding acquisition, Resources, Writing – review & editing

Alexander Jung: Conceptualization, Data curation, Funding acquisition, Formal analysis, Investigation, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing

Appendices

A AP biomarkers and abnormalities

The AP biomarkers used in this study were selected such that the key characteristics of the depolarization and the repolarization phase can be quantified. They include *RMP* (resting transmembrane potential measured just before stimulation), *dV mMax* (maximum transmembrane potential slope during the upstroke), *Peak* (peak transmembrane potential at the end of the upstroke), *APD_x* (AP duration at $x \in \{30, 40, 50, 60, 70, 80, 90\}$ % repolarization relative to the AP amplitude (*Peak* – *RMP*) measured from the instant of *dV mMax*), and *Tri₉₀₋₄₀* (triangulation defined as the difference between *APD₉₀* and *APD₄₀* [Britton et al., 2017]).

For the GSA (see Appendix B) and the creation of synthetic data (see Section 2.3.1), AP abnormalities were also considered which included depolarization abnormalities, repolarization abnormalities, and alternans. Depolarization abnormalities were defined as an upstroke peak below 0 mV and an AP that does not reach 0 mV before 100 ms after stimulation [Passini et al., 2017]. Repolarization abnormalities were defined as a transmembrane potential rate of rise of more than $0.01 \frac{\text{mV}}{\text{ms}}$ from 150 ms after the upstroke peak onwards (representative of early afterdepolarizations) and as a transmembrane potential that does not fall below –40 mV [Passini et al., 2017] (representative of repolarization failure). Alternans were defined as *APD₉₀* difference of more than 5 ms between two consecutive APs [Morotti et al., 2021].

B Global sensitivity analysis

A variance-based Sobol' global sensitivity analysis (GSA) [Sobol', 2001] was performed on the simulator to quantify the sensitivities of the maximum channel conductances (inputs) with respect to the AP biomarkers (outputs; see Appendix A). This informed the decision on which inputs to consider in the emulator. Furthermore, it was used for the interpretation of the solutions of the inverse problem since parameters that are insensitive with respect to the outputs indicate non-identifiability [Guillaume et al., 2019].

The maximum conductances used for building the model population in [Tomek et al., 2019 [DOI](#)] were considered and Saltelli's sampling scheme [Saltelli, 2002 [DOI](#)] was applied with $N = 1024$ to generate 20,480 input samples with values between 50% and 150% of the original values. Simulations were performed for each input sample and biomarker values (see Appendix A) derived from the last AP were used for the analysis. However, data were excluded if not all biomarkers could be determined or abnormalities (see Appendix A) were detected in the last two consecutive APs. First-order (S1) and total-effect (ST) Sobol' sensitivity indices were computed using the Saltelli method [Homma and Saltelli, 1996 [DOI](#), Saltelli, 2002 [DOI](#)]. This requires outputs for each input sample and to take this into account, excluded outputs were assigned the mean values of included outputs. The SALib-Sensitivity Analysis Library [Herman and Usher, 2017 [DOI](#)] was used for the GSA.

The GSA could be performed on the data of all input samples as no data were excluded. S1 and ST were mostly very similar which indicates only little interactions among the maximum conductivities relative to the AP biomarkers (**Figure 11** [DOI](#)). The only exception was *TP*. As was to be expected, the analysis underlines the predominant relative sensitivity of G_{Na} with respect to biomarkers of the depolarization phase, the predominant relative sensitivity of G_{NaL} , G_{Kr} , and G_{NCX} with respect to biomarkers of the repolarization phase and the predominant relative sensitivity of G_{K1} , P_{NaK} to the resting transmembrane potential. However, G_{Ks} has a negligible relative sensitivity to all biomarkers. This indicates almost non-identifiability.

C EAD classification

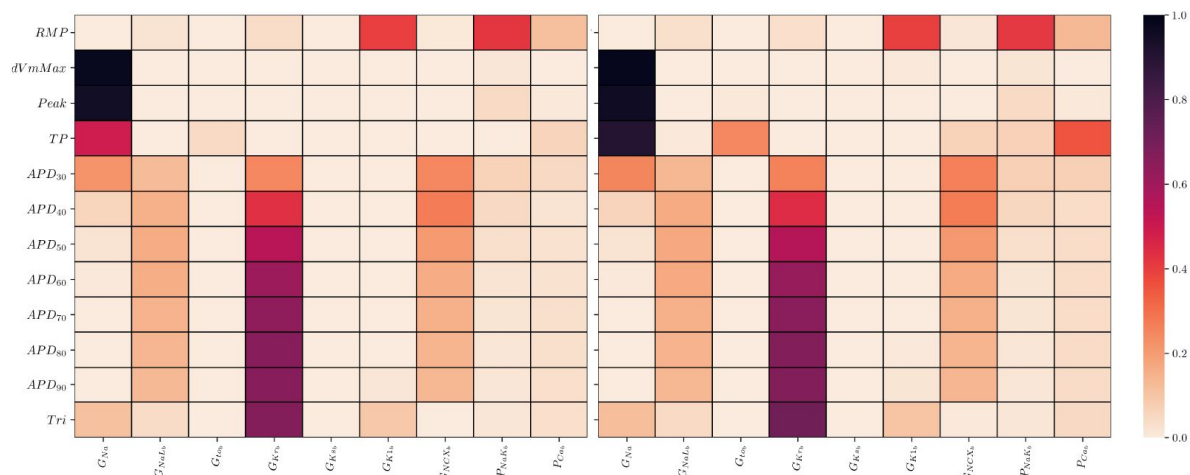


Figure 11

Global sensitivity analysis of the ToR-ORd simulator. Sobol' sensitivity indices are shown for each maximum conductance relative to each AP biomarker. Left: first-order (S1), right: total-effect (ST) Sobol' sensitivity coefficient.

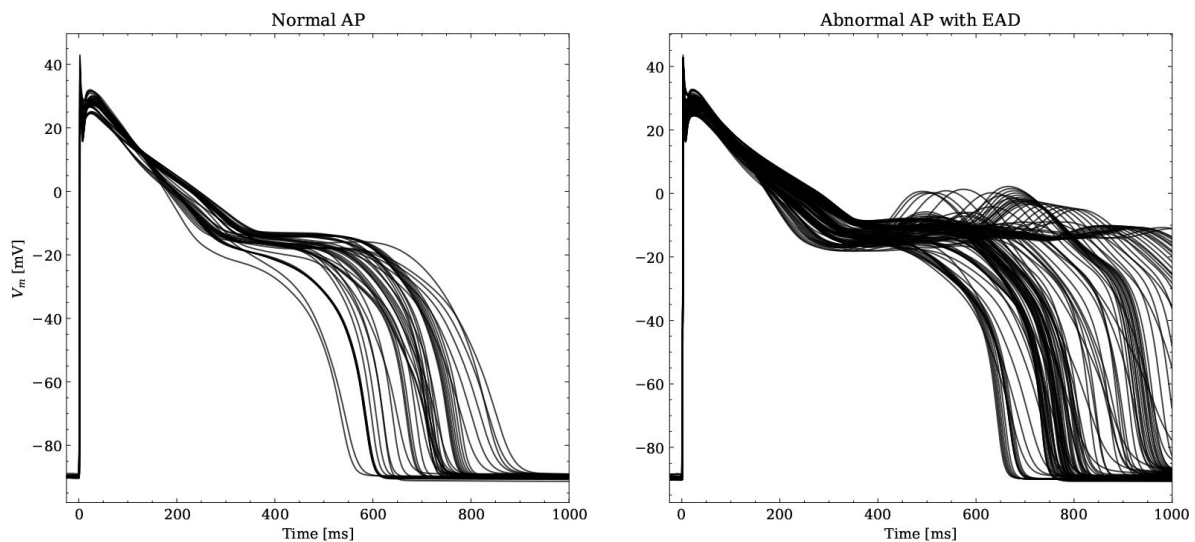


Figure 12

Emulated APs based on the pharmacological parameters of data set #2. See also [Section 3.1.1](#) and [Figure 7](#). Left: APs classified as normal, right: APs classified as abnormal exhibiting EADs. The classification criterion is outlined in Appendix A.

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Reviewer #1 (Public Review):

Summary:

The authors present a neural network (NN)-based approach to computationally cheaper emulation of simulations of biophysically relatively detailed cardiac cell models based on systems of ordinary differential equations. Relevant case studies are used to demonstrate the performance in the prediction of standard action potentials, as well as action potentials

manifesting early depolarizations. Application to the "reverse problem" (inferring the effect of pharmacological compounds on ion channels based on action potential data before and after drug treatment) is also explored, which is a task of generally high interest.

Strengths:

This is a well-designed study, which explores an area that many in the cardiac simulation community will be interested in. The article is well written and I particularly commend the authors on transparency of methods description, code sharing, etc. - it feels rather exemplary in this regard and I only wish more authors of cardiac simulation studies took such an approach. The training speed of the network is encouraging and the technique is accessible to anyone with a reasonably strong GPU, not needing specialized equipment.

Weaknesses:

Below are several points that I consider to be weaknesses and/or uncertainties of the work:

1. I am not convinced by the authors' premise that there is a great need for further acceleration of cellular cardiac simulations - it is easy to simulate tens of thousands of cells per day on a workstation computer, using simulation conditions similar to those of the authors. I do not really see an unsolved task in the field that would require further speedup of single-cell simulations.

At the same time, simulations offer multiple advantages, such as the possibility to dissect mechanisms of the model behaviour, and the capability to test its behaviour in a wide array of protocols - whereas a NN is trained for a single purpose/protocol, and does not enable a deep investigation of mechanisms. Therefore, I am not sure the cost/benefit ratio is that strong for single-cell emulation currently.

An area that is definitely in need of acceleration is simulations of whole ventricles or hearts, but it is not clear how much potential for speedup the presented technology would bring there. I can imagine interesting applications of rapid emulation in such a setting, some of which could be hybrid in nature (e.g. using simulation for the region around the wavefront of propagating electrical waves, while emulating the rest of the tissue, which is behaving more regularly/predictable, and is likely to be emulated well), but this is definitely beyond of the scope of this article.

2. The authors run a cell simulation for 1000 beats, training the NN emulator to mimic the last beat. It is reported that the simulation of a single cell takes 293 seconds, while emulation takes only milliseconds, implying a massive speedup. However, I consider the claimed speedup achieved by emulation to be highly context-dependent, and somewhat too flattering to the presented method of emulation. Two specific points below:

First, it appears that a not overly efficient (fixed-step) numerical solver scheme is used for the simulation. On my (comparable, also a Threadripper) CPU, using the same model ("ToR-ORDyndncl"), but a variable step solver ode15s in Matlab, a simulation of a cell for 1000 beats takes ca. 50 seconds, rather than 293 of the authors. This can be further sped up by parallelization when more cells than available cores are simulated: on 32 cores, this translates into ca. 2 seconds amortized time per cell simulation (I suspect that the NN-based approach cannot be parallelized in a similar way?). By amortization, I mean that if 32 models can be simulated at once, a simulation of X cells will not take $X \cdot 50$ seconds, but $(X/32) \cdot 50$. (with only minor overhead, as this task scales well across cores).

Second, and this is perhaps more important - the reported speed-up critically depends on the number of beats in the simulation - if I am reading the article correctly, the runtime compares a simulation of 1000 beats versus the emulation of a single beat. If I run a simulation of a single beat across multiple simulated cells (on a 32-core machine), the amortized runtime is around 20 ms per cell, which is only marginally slower than the NN emulation. On the other

hand, if the model was simulated for aeons, comparing this to a fixed runtime of the NN, one can get an arbitrarily high speedup.

Therefore, I'd probably emphasize the concrete speedup less in an abstract and I'd provide some background on the speedup calculation such as above, so that the readers understand the context-dependence. That said, I do think that a simulation for anywhere between 250 and 1000 beats is among the most reasonable points of comparison (long enough for reasonable stability, but not too long to beat an already stable horse; pun with stables was actually completely unintended, but here it is...). I.e., the speedup observed is still valuable and valid, albeit in (I believe) a somewhat limited sense.

3. It appears that the accuracy of emulation drops off relatively sharply with increasing real-world applicability/relevance of the tasks it is applied to. That said, the authors are to be commended on declaring this transparently, rather than withholding such analyses. I particularly enjoyed the discussion of the not-always-amazing results of the inverse problem on the experimental data. The point on low parameter identifiability is an important one and serves as a warning against overconfidence in our ability to infer cellular parameters from action potentials alone. On the other hand, I'm not that sure the difference between small tissue preps and single cells which authors propose as another source of the discrepancy will be that vast beyond the AP peak potential (probably much of the tissue prep is affected by the pacing electrode?), but that is a subjective view only. The influence of coupling could be checked if the simulated data were generated from 2D tissue samples/fibres, e.g. using the Myokit software.

Given the points above (particularly the uncertain need for further speedup compared to running single-cell simulations), I am not sure that the technology generated will be that broadly adopted in the near future. However, this does not make the study uninteresting in the slightest - on the contrary, it explores something that many of us are thinking about, and it is likely to stimulate further development in the direction of computationally efficient emulation of relatively complex simulations.

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Reviewer #2 (Public Review):

Summary:

This study provided a neural network emulator of the human ventricular cardiomyocyte action potential. The inputs are the corresponding maximum conductances and the output is the action potential (AP). It used the forward and inverse problems to evaluate the model. The forward problem was solved for synthetic data, while the inverse problem was solved for both synthetic and experimental data. The NN emulator tool enables the acceleration of simulations, maintains high accuracy in modeling APs, effectively handles experimental data, and enhances the overall efficiency of pharmacological studies. This, in turn, has the potential to advance drug development and safety assessment in the field of cardiac electrophysiology.

Strengths:

(1) Low computational cost: The NN emulator demonstrated a massive speed-up of more than 10,000 times compared to the simulator. This substantial increase in computational speed has the potential to expedite research and drug development processes

(2) High accuracy in the forward problem: The NN emulator exhibited high accuracy in solving the forward problem when tested with synthetic data. It accurately predicted normal APs and, to a large extent, abnormal APs with early afterdepolarizations (EADs). High accuracy is a notable advantage over existing emulation methods, as it ensures reliable modeling and prediction of AP behavior

Weaknesses:

(1) Input space constraints: The emulator relies on maximum conductances as inputs, which explain a significant portion of the AP variability between cardiomyocytes. Expanding the input space to include channel kinetics parameters might be challenging when solving the inverse problem with only AP data available.

(2) Simplified drug-target interaction: In reality, drug interactions can be time-, voltage-, and channel state-dependent, requiring more complex models with multiple parameters compared to the oversimplified model that represents the drug-target interactions by scaling the maximum conductance at control. The complex model could also pose challenges when solving the inverse problem using only AP data.

(3) Limited data variety: The inverse problem was solved using AP data obtained from a single stimulation protocol, potentially limiting the accuracy of parameter estimates. Including AP data from various stimulation protocols and incorporating pacing cycle length as an additional input could improve parameter identifiability and the accuracy of predictions.

(4) Larger inaccuracies in the inverse problem using experimental data: The reasons for this result are not quite clear. Hypotheses suggest that it may be attributed to the low parameter identifiability or the training data set were collected in small tissue preparation.

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Reviewer #3 (Public Review):

Summary:

Grandits and colleagues were trying to develop a new tool to accelerate pharmacological studies by using neural networks to emulate the human ventricular cardiomyocyte action potential (AP). The AP is a complex electrical signal that governs the heartbeat, and it is important to accurately model the effects of drugs on the AP to assess their safety and efficacy. Traditional biophysical simulations of the AP are computationally expensive and time-consuming. The authors hypothesized that neural network emulators could be trained to predict the AP with high accuracy and that these emulators could also be used to quickly and accurately predict the effects of drugs on the AP.

Strengths:

One of the study's major strengths is that the authors use a large and high-quality dataset to train their neural network emulator. The dataset includes a wide range of APs, including normal and abnormal APs exhibiting EADs. This ensures that the emulator is robust and can be used to predict the AP for a variety of different conditions.

Another major strength of the study is that the authors demonstrate that their neural network emulator can be used to accelerate pharmacological studies. For example, they use the emulator to predict the effects of a set of known arrhythmogenic drugs on the AP. The emulator is able to predict the effects of these drugs, even though it had not been trained on these drugs specifically.

Weaknesses:

One weakness of the study is that it is important to validate neural network emulators against experimental data to ensure that they are accurate and reliable. The authors do this to some extent, but further validation would be beneficial. In particular for the inverse problem, where the estimation of pharmacological parameters was very challenging and led to particularly large inaccuracies.

Additional context:

The work by Grandits et al. has the potential to revolutionize the way that pharmacological studies are conducted. Neural network emulation has the promise to reduce the time and cost of drug development and to improve the safety and efficacy of new drugs. The methods and data presented in the paper are useful to the community because they provide a starting point for other researchers to develop and improve neural network emulators for the human ventricular cardiomyocyte AP. The authors have made their code and data publicly available, which will facilitate further research in this area.

It is important to note that neural network emulation is still a relatively new approach, and there are some challenges that need to be addressed before it can be widely adopted in the pharmaceutical industry. For example, neural network emulators need to be trained on large and high-quality datasets. Additionally, it is important to validate neural network emulators against experimental data to ensure that they are accurate and reliable. Despite these challenges, the potential benefits of neural network emulation for pharmacological studies are significant. As neural network emulation technology continues to develop, it is likely to become a valuable tool for drug discovery and development.

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